

AMDERMA, Russian FederationWMO: **230220**Lat: **69.750N**Lon: **61.700E**Elev: **49**StdP: **100.74**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.4	-31.2	-35.8	0.1	-33.2	-33.9	0.2	-31.1	19.5	-14.4	17.7	-14.9	4.2	90	1.037

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	20.4	16.1	17.4	14.3	14.6	12.5	16.5	19.8	14.5	17.0	12.5	14.6	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	10.7	18.5	13.0	9.4	15.9	11.1	8.3	13.9	46.4	19.7	40.8	17.2	35.7	14.7	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.5	13.0	DB	-34.9	25.3	3.5	3.0	-37.5	27.5	-39.5	29.2	-41.5	30.9	-44.0	33.1	(4)
(5)			WB	-35.2	19.1	3.3	2.1	-37.6	20.6	-39.5	21.8	-41.4	23.0	-43.8	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.3	-17.2	-17.5	-14.6	-10.0	-4.0	3.0	7.8	7.8	4.3	-1.8	-9.3	-13.2	(6)
(7)		DBStd	10.89	7.69	8.34	7.67	6.40	4.62	4.45	4.88	3.13	3.09	4.55	6.11	7.79	(7)
(8)		HDD10.0	5660	844	770	762	601	433	221	104	86	173	364	580	721	(8)
(9)		HDD18.3	8639	1102	1003	1021	851	692	462	328	328	421	623	830	979	(9)
(10)		CDD10.0	65	0	0	0	0	0	9	37	17	2	0	0	0	(10)
(11)		CDD18.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(11)
(12)		CDH23.3	21	0	0	0	0	0	3	17	1	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.3	7.3	7.2	6.6	6.5	5.8	5.2	4.7	5.1	5.5	6.4	7.2	7.5	(14)
(15)	Precipitation	PrecAvg	398	25	19	19	19	26	35	50	49	46	46	33	29	(15)
(16)		PrecMax	509	62	51	63	38	48	76	110	91	123	75	86	53	(16)
(17)		PrecMin	254	1	0	4	0	5	9	8	18	17	17	7	6	(17)
(18)		PrecStd	62	17	12	11	10	12	19	33	19	25	15	17	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.6	-0.8	0.7	3.0	11.1	23.0	25.6	20.5	15.5	7.9	1.3	0.3	(19)
(20)			MCWB	-1.0	-1.1	-0.1	1.8	8.3	15.9	18.8	16.1	12.3	7.0	0.9	0.0	(20)
(21)		2%	DB	-2.1	-2.7	-0.8	1.2	5.7	17.5	21.9	17.5	12.1	6.1	0.5	-0.5	(21)
(22)			MCWB	-2.4	-3.0	-1.2	0.5	4.3	13.6	17.3	14.6	10.4	5.5	0.1	-0.9	(22)
(23)		5%	DB	-4.0	-4.1	-2.0	0.2	3.3	13.4	19.1	15.1	10.3	4.8	-0.3	-1.7	(23)
(24)			MCWB	-4.4	-4.4	-2.3	-0.3	2.2	10.7	15.5	13.0	9.2	4.2	-0.7	-2.0	(24)
(25)		10%	DB	-6.3	-6.1	-3.5	-1.0	1.6	9.7	16.0	12.9	8.8	3.6	-1.6	-2.8	(25)
(26)			MCWB	-6.7	-6.5	-3.9	-1.4	1.0	7.9	13.6	11.5	7.9	3.0	-2.0	-3.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	-1.1	0.1	1.9	8.5	16.2	19.7	16.9	13.1	7.1	1.1	0.2	(27)
(28)			MCDB	-0.8	-0.9	0.5	2.8	10.9	21.2	24.4	19.5	15.2	7.6	1.2	0.2	(28)
(29)		2%	WB	-2.4	-2.9	-1.1	0.7	4.5	13.8	17.8	14.7	10.6	5.6	0.3	-0.8	(29)
(30)			MCDB	-2.1	-2.7	-0.9	1.1	5.6	17.3	21.2	17.1	11.7	6.1	0.5	-0.6	(30)
(31)		5%	WB	-4.4	-4.5	-2.2	-0.2	2.3	10.6	15.8	13.2	9.3	4.2	-0.6	-2.0	(31)
(32)			MCDB	-4.1	-4.2	-1.9	0.1	3.1	13.3	18.8	15.0	10.2	4.7	-0.4	-1.7	(32)
(33)		10%	WB	-6.7	-6.5	-3.9	-1.4	1.0	8.0	13.6	11.5	8.0	3.1	-1.9	-3.2	(33)
(34)			MCDB	-6.3	-6.1	-3.5	-1.0	1.6	9.6	15.9	12.8	8.8	3.5	-1.6	-2.9	(34)
(35)	Mean Daily Temperature Range	MDBR	6.1	6.0	6.1	6.2	4.5	4.9	6.0	4.6	4.1	3.8	5.4	5.9	(35)	
(36)		5% DB	MCDBR	7.0	6.1	4.9	5.7	6.4	12.2	12.0	9.5	6.6	3.8	5.0	5.3	(36)
(37)			MCWBR	7.1	6.1	4.9	5.5	5.3	8.5	8.0	6.8	5.1	3.6	5.1	5.3	(37)
(38)		5% WB	MCDBR	6.8	5.9	5.0	5.5	6.1	11.9	11.4	8.8	6.1	3.6	5.0	5.3	(38)
(39)			MCWBR	6.9	6.0	5.0	5.4	5.2	8.6	8.0	6.7	4.9	3.5	5.2	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.210	0.241	0.292	0.319	0.334	0.339	0.372	0.353	0.315	0.258	N/A	N/A	(40)	
(41)		taud	1.874	1.965	2.036	2.061	2.089	2.289	2.362	2.428	2.457	2.164	N/A	N/A	(41)	
(42)		Ebn at Noon	11	462	690	800	836	851	804	777	707	454	0	0	(42)	
(43)		Edh at Noon	3	45	85	115	127	108	98	82	59	36	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.01	0.30	1.76	3.84	5.46	5.43	4.91	3.02	1.56	0.40	0.00	0.00	(44)	
(45)		RadStd	0.00	0.07	0.20	0.31	0.33	0.55	0.67	0.29	0.13	0.06	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+1.72	+2.78	+3.36	+1.64	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (3 neighbors)	+1.72	+2.78	+3.36	+1.64	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page